

Policy Shocks: Stock Market Response and
Financial Impact of Indonesia's Nickel Export Ban

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Abstract

Indonesia is the world's largest producer of nickel, accounting for 40% of global production, with reserves exceeding 21 million metric tons. In January 2020, the Indonesian government put in place an export ban on raw nickel ores. This was intended to promote the downstreaming of nickel into value-added industries such as electric vehicle battery production and stainless-steel manufacturing. This study explores the causal impact of the export ban announcement on the stock market valuation of nickel firms in Indonesia. To isolate the effect of the export ban announcement, we employ an event study methodology to compare the stock returns of Indonesian nickel producers and consumers (treatment groups) against palm oil firms (control group).

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1 Introduction

On January 1, 2020, Indonesia banned the export of its most valuable commodity: nickel. The government had decided that the future global economy will rely on nickel to fuel the energy transition, and it had decided to leverage its overwhelming control of world nickel to propel Indonesian industrialization. This paper explores the effects of Indonesia’s 2020 nickel export ban on the stock prices of nickel-related corporations. Specifically, we ask: How does the export ban put in place by the Indonesian government impact the market valuation of domestic nickel companies, and does this impact differ between firms that extract raw nickel versus those that process it into value-added products?

Indeed, this export ban has been effective in pushing for nickel downstreaming in Indonesia, evident from the increase of export-generated revenue from USD 6 billion in 2013 to USD 30 billion in 2022 and the birth of new Indonesian corporations or subsidiaries that process nickel into its derivatives (Abdurrachman, 2023). This increase in operational performance and profitability would naturally translate to higher stock valuations for such companies. However, there are other potential drivers of stock prices that coincided with the implementation of this export ban (e.g., the expanding electric-vehicle market, the fluctuating global nickel prices, and the post-pandemic global economic rebound). This study employs an event study methodology to isolate the causal effect of the policy shock on firm value, as reflected in stock market returns.

Numerous papers have delved into the topic of trade restrictions and their effects on the economy. Furthermore, there is an abundance of literature discussing the over-reliance of the Indonesian economy on unprocessed natural resource exports, with some even focusing on nickel specifically. The existing literature primarily offers theoretical and qualitative discussions on the impact of policies related to Indonesia’s nickel industry. To our best knowledge, this study is the first to employ stock market data in an event study framework to analyze the effects of Indonesia’s 2020 nickel export ban, thereby contributing to the

understanding of resource export policies and their role in fostering domestic value-added industries.

1.1 Institutional Background: Ministry of Energy, Regulation 11/2019

Indonesia is not only the world’s leading producer of nickel, accounting for 40% of global production, it holds 21 million metric tons of nickel reserves (Mine, 2023)—enough to power the global green energy transition towards electric vehicles for the decades to come. The Indonesian economy has long relied on raw material exports to drive GDP growth. This phenomenon, commonly known as Dutch Disease, has hindered the country’s ability to develop its domestic value-adding industries (Raidi, 2024). To combat this, the Jokowi administration issued an export ban on raw nickel ores in January of 2020.

In late 2019, the Ministry of Energy and Mineral Resources issued Regulation Number 11/2019, which effectively ended all permits for the export of raw nickel ores by December 31st, 2019, throughout Indonesia (Ministry of Energy and Mineral Resources, 2019). This policy change by the government seeks to incentivize the development of key domestic nickel processing industries, such as in electric vehicle batteries, energy storage systems, super-alloys for defense, and stainless-steel production, all of which have high growth potentials (Yanwardhana, 2023). The regulation represents an exogenous policy shock that provides a natural experiment for examining how trade restrictions affect firm value across different segments of the nickel supply chain.

2 Literature Review

Our study is related to literature on export-import policy and the nickel industry, a central topic in trade and developmental economics research. These disciplines are especially significant in the increasingly protectionist global economy, where developing countries are facing greater obstacles in escaping the middle-income trap and achieving developed economy

status. Extensive research has been produced in these fields.

For instance, in the paper "Economics of Export Restrictions as Applied to Industrial Raw Materials", K. Fung and J. Korinek (2013) provides foundational insights into the broader macroeconomic implications of implementing export restrictions on key commodities. Such policies include the use of taxes, quotas, and outright bans as well, which is pertinent to our discussion of Regulation 11/2019. They find that export quotas, which more closely resembles a ban, has far more wide-reaching effects than an export tax on global prices and profits. This occurs since firms may not increase their levels of production when a quota is imposed upon them as opposed to a tax, leading to a sharper rise in price when demand increases. This concept is particularly relevant to our case study on how markets value firms affected by such restrictions. Although quotas are less strict than the ban imposed by Regulation 11/2019, it may have similar consequences to the Indonesian nickel industry that our paper attempts to study.

Similarly, Andarini et al. (2024) also finds bans and quotas to be much more effective in the development of the national economy, particularly in Indonesia, where capital-intensive mining industries play a key role. However, this study adopts a theoretical and policy-focused approach, rather than the quantitative analysis of policy effects that our paper attempts to provide. Our paper advances the literature through our use of empirically driven methods to quantify the effects of such export bans on firm value, particularly in the nickel industry. By using stock market data rather than accounting-based measures, we capture forward-looking investor expectations about policy impacts. Our quantitative study, along with existing qualitative and theoretical analysis on export bans, strengthens the case for exploring the effects of export restriction on the development of the national economy. Frankel (2010) notes that many resource-rich countries, like Indonesia, face the "Natural Resource Curse", a paradoxical phenomenon that causes such states to experience lower economic growth. The paper highlights how the abundance of resources tend to result in certain conditions being formed within the economy, such as commodity price volatility, the lack of development

in other sectors, and poor institutional governance. Our study narrows down the focus of this topic by examining whether an export ban can be considered a remedy to this 'curse', especially given that the characteristics of a country with the 'natural resource curse' apply to the Indonesian economy.

Hadad et. al (2022) conducts a legal analysis on the export restriction we are studying (i.e. Regulation 11/2019) with respect to international trade laws established by the World Trade Organization and the General Agreement on Tariffs and Trade (GATT). The legal perspective the authors take presumes the benefit of such a policy in fostering domestic industries but argues that the legal challenges posed may become a detriment to the economy. This paper complements our study by providing crucial context on the regulatory and legal framework of Regulation 11/2019: the policy shock we analyze through stock market reactions. Whereas this paper lays out the legal context surrounding the export ban, our study establishes a causal relationship and quantifies the effect of this ban on the market valuation of nickel companies.

Pandyaswargo et al. (2021) provides a SWOT analysis and an analysis on the industry positioning of the electric vehicle industry in Indonesia in relation to the nickel export ban, which complements well with our analysis of how markets valued the nickel sector's prospects as it provides valuable context into the current industry landscape. We will utilize concrete econometric approaches, specifically event study methodology, to measure the market's assessment of the export ban's efficacy that this paper argues qualitatively.

Within the literature on export policy and the Indonesian nickel industry, we contribute, to our knowledge, the first evaluation using stock market data in an event study framework to measure the causal effect of the Ministry of Energy and Mineral Resources Regulation Number 11/2019 on the market valuation of companies in the domestic nickel sector in Indonesia. While previous studies rely on aggregate trade statistics, firm-level financial statements, or theoretical and qualitative arguments, we utilize publicly accessible market data to capture forward-looking investor expectations and perceptions about national-level policy impacts

on individual nickel firms, using event study methodology to establish causality.

3 Data

3.1 Data Collection and Sources

To effectively implement our event study analysis, we have identified concrete data sources to build a robust empirical framework. We are interested in the daily stock returns of nickel and palm oil companies, both of which represent key sectors for the Indonesian economy between January 2018 and February 2020. On September 2, 2019, the Ministry of Energy and Mineral Resources announced that nickel ore exports would be banned effective January 2020 (Ministry of Energy and Mineral Resources, 2019). We used data directly accessible on the Ministry of Energy and Mineral Resources website (Ministry of Energy and Mineral Resources, 2020), the Indonesian Stock Exchange website (Indonesian Stock Exchange, n.d.), and the Association of Indonesian Nickel Miners website (APNI, n.d.) to create a comprehensive list of public nickel companies. For palm oil companies, we use the Indonesian Palm Oil Association website (GAPKI, n.d.) and other similar sources. To further isolate the treatment effect, we use the Jakarta Composite Index as our market benchmark to control for overall market movements that could confound the relationship between nickel company stock returns and the export ban announcement. Daily stock prices for all corporations and the Jakarta Composite Index can be found on Bloomberg Terminal (Bloomberg L.P., 2025).

We explicitly specify what the data we use for the analysis is. The data spans from January 2, 2018 to February 19, 2020. Our primary outcome of interest is represented by $R_{i,t}$, which is the daily stock return for company i at time t . The treatment group dataset consists of 8 different nickel companies across the sample period. Of these 8 nickel companies, 5 primarily focus on the extraction and production of raw nickel ores (producers). The other 3 companies are firms where their primary business model relies on the consumption and processing of nickel into value-added products such as stainless steel and aluminum

(consumers).

Similarly, the control group dataset consists of 7 different palm oil companies across the sample period. In total, there are 15 publicly traded companies with sufficient data quality for our event study methodology, covering an estimation window of 400 trading days (January 2, 2018 to September 1, 2019) and an event window of 118 trading days (September 2, 2019 to February 19, 2020).

Table 1: Sample Companies by Treatment Group

Ticker	Company Name	Company Group
INCO IJ Equity	PT Vale Indonesia Tbk	Nickel Producer
ANTM IJ Equity	PT Aneka Tambang Tbk	Nickel Producer
DKFT IJ Equity	PT Central Omega Resources Tbk	Nickel Producer
CITA IJ Equity	PT Cita Mineral Investindo Tbk	Nickel Producer
ZINC IJ Equity	PT Kapuas Prima Coal Tbk	Nickel Producer
HRUM IJ Equity	PT Harum Energy Tbk	Nickel Consumer
INAI IJ Equity	PT Indal Aluminium Industry Tbk	Nickel Consumer
GDST IJ Equity	PT Gunawan Dianjaya Steel Tbk	Nickel Consumer
AALI IJ Equity	PT Astra Agro Lestari Tbk	Palm Oil (Control)
SMAR IJ Equity	PT Sinar Mas Agro Resources and Technology Tbk	Palm Oil (Control)
LSIP IJ Equity	PT PP London Sumatra Indonesia Tbk	Palm Oil (Control)
SIMP IJ Equity	PT Salim Ivomas Pratama Tbk	Palm Oil (Control)
TBLA IJ Equity	PT Tunas Baru Lampung Tbk	Palm Oil (Control)
SGRO IJ Equity	PT Sampoerna Agro Tbk	Palm Oil (Control)
DSNG IJ Equity	PT Dharma Satya Nusantara Tbk	Palm Oil (Control)

In addition to the daily stock returns of all these companies, we collect data on the Jakarta Composite Index returns between January 2018 and February 2020, given that overall market movements have a high likelihood of impacting the stock returns of our companies of interest. This variable helps isolate the causal effect of the nickel export ban

announcement on abnormal stock returns through the market model framework. We obtain all stock price and index data from Bloomberg Terminal (Bloomberg L.P., 2025), which provides comprehensive and reliable daily trading data for Indonesian public companies.

3.2 Data Collection Challenges

One of the primary challenges we face in data collection is that most nickel companies in Indonesia engage in both raw nickel ore extraction and nickel ore processing operations. Therefore, it is arguably difficult to determine whether to categorize a company as a nickel producer or a nickel consumer. For instance, the State-Owned Enterprise (SOE) PT. Aneka Tambang Tbk. is a corporation that engages in both types of operations. Furthermore, their business activities span multiple segments, making it difficult to cleanly separate exposure to raw nickel mining versus nickel processing. We address this problem by qualitatively assessing whether the primary business activity of a corporation is nickel extraction or processing based on revenue composition disclosed in annual reports and company descriptions. Thereafter, we categorize each firm as either a producer or consumer for the purposes of our event study analysis.

Another challenge in data collection is the prevalence of missing data in the reported daily stock prices, even on Bloomberg Terminal. Given the nature of emerging markets such as Indonesia, the daily stock prices of medium to small cap companies are not as consistently recorded by stock indexes, resulting in fewer small cap nickel companies having sufficient data available, especially for the earlier years of our estimation window. As a consequence, some otherwise informative nickel firms had to be removed from the study given the high percentage of missing daily stock prices. For instance, LION IJ Equity was excluded from our analysis due to 41% missing data in the event window, as this level of missingness would compromise the reliability of our abnormal return estimates.

3.3 Summary Statistics

Table 2: Summary Statistics by Company Group

Company_Group	N_companies	Mean_return	SD_return
Nickel Producer	5	0.0818	3.3336
Nickel Consumer	3	-0.0182	3.7135
Palm Oil (Control)	7	-0.0519	2.4509

The time series summary statistics shown in Table 2 shows the mean returns and standard deviation of said mean returns for each firm group. As shown, among the 5 nickel producing firms, they have an average of 0.0818 in daily stock returns, with a standard deviation of 3.3336. In contrast, both the nickel consumer firms, which number 3 firms in total, and the palm oil firms, which number 7 firms in total, have a negative average daily stock returns of -0.0182 and -0.0519, respectively, and a standard deviation of 3.7135 and 2.4509, respectively.

Table 3: Additional Summary Statistics (Thousands USD for Profit)

Commodity	Measure	Mean	SD	Median	Min	Max	Size
Nickel	Producer Profit	56,066	101,084	18,450	-4,400	541,500	56
	Consumer Profit	141,786	151,925	83,600	0	553,300	56
	Price (USD/Ton)	16	6	14	10	27	8
Palm Oil	Profit	416,964	1,223,133	91,400	-16,900	7,191,200	128
	Price (USD/Ton)	0.8	0.2	0.7	0.4	1	8

While our event study methodology relies on daily stock returns rather than accounting profits, we present industry-level comparability metrics in Table 3 to provide economic con-

text for our control group selection. The average profit earned by both nickel extractors and nickel value-adders are comparable in magnitude to the average profit earned by palm oil companies, whereby both are between the tens of millions to hundreds of millions of dollars. Nickel prices have a standard deviation (SD) of USD 6,000/ton, while palm oil prices have an SD of USD 200/ton.

At first glance, this disparity in price volatility between the two commodities, as captured by the standard deviation, may suggest that palm oil may not be as suitable a control group. However, we believe that this does not disqualify palm oil as the absolute dollar price volatility of these two commodities become comparable when adjusted for the much larger volumes that palm oil is traded at relative to nickel ores. Specifically, nickel production reached 3.4 million tons, which means their aggregate annual price volatility is $3,400,000 * 6,000 = \text{USD } 20.4 \text{ billion}$ (Nornickel, 2023), whereas palm oil production reached 80 million tons, which means their aggregate annual price volatility is $80,000,000 * 200 = \text{USD } 16 \text{ billion}$ (Statista, 2024). As such, palm oil may be considered an appropriate export-oriented control group given the summary statistics shown in Table 3.

This volatility in price for nickel may be due to several factors: the continually fluctuating demand for steel and batteries due to the speculative nature of the electric vehicle market, the stockpiling activities of many economies with regards to nickel ore, and the export ban policy we are studying. Similarly, the price volatility of palm oil can be attributed to the fluctuating demand for edible oils and fuels, the uncertain supply of labor and fertilizers, changing weather conditions, and trade restrictions.

Table 4: Data Quality Metrics (Event Window: September 2, 2019 - February 19, 2020)

Ticker	Company Group	Valid Days	Missing %
ANTM IJ Equity	Nickel Producer	118	0.00
CITA IJ Equity	Nickel Producer	118	0.00
DKFT IJ Equity	Nickel Producer	118	0.00
INCO IJ Equity	Nickel Producer	118	0.00
ZINC IJ Equity	Nickel Producer	118	0.00
GDST IJ Equity	Nickel Consumer	118	0.00
HRUM IJ Equity	Nickel Consumer	118	0.00
INAI IJ Equity	Nickel Consumer	107	9.32
AALI IJ Equity	Palm Oil (Control)	118	0.00
DSNG IJ Equity	Palm Oil (Control)	118	0.00
LSIP IJ Equity	Palm Oil (Control)	118	0.00
SGRO IJ Equity	Palm Oil (Control)	101	14.41
SIMP IJ Equity	Palm Oil (Control)	118	0.00
SMAR IJ Equity	Palm Oil (Control)	109	7.63
TBLA IJ Equity	Palm Oil (Control)	118	0.00
<i>Note: LION IJ Equity excluded due to 41% missing data in original event window.</i>			

Due to the nature of Indonesia being an emerging stock market, many otherwise viable and suitable firms were removed from this study due to the lack of availability of daily stock price data. Based on extracting data from Bloomberg, any firm with more than 20% missing data within the estimation window and event window were excluded from any analysis, resulting in the displayed firms being included.

4 Methodology

To explore the data, we will use R to conduct event study analysis and generate the necessary variables for our regression models. The dataset is designed to achieve fine granularity with daily stock price observations. It is structured such that each row represents the daily trading data for a single company, including columns for `Company_ID`, `Company_Group` (indicating whether the company is a nickel producer, nickel consumer, or palm oil control), `Daily_Return`, `Market_Return`, `Date`, and `Post` (indicating whether the period is post-announcement).

We will interpret the coefficients of interest (i.e., γ_1 and γ_2 from Model 4). Thereafter, we will assess the statistical significance of the estimated coefficients from our cross-sectional regression model at conventional significance levels to determine whether Regulation 11/2019, the nickel export ban announcement, has a causal effect on the cumulative abnormal returns of nickel companies in Indonesia.

4.1 Event Study

One of our main methodology assumptions is the efficient market hypothesis, which requires that Indonesian capital markets have the ability to absorb new information quickly and adjust prices accordingly. This information must also be exogenous and unexpected to investors. This is represented by the announcement date of the regulation (September 2nd, 2019) rather than the date where the policy takes into effect (January 1st, 2020). As such, we design our regression models to be centered around that date of announcement, whereby our estimation window ends and our event window begins, respectively.

Additionally, another primary assumption would be the lack of a confounding event that may impact investor behavior towards the security within the window of study. Initially, we designed the study to be symmetric between the estimation window and event window. However, the significance of the coefficients in earlier models were lacking, which we attributed

to the unaccounted impact of the outbreak of COVID-19. We found that stocks and investor behavior worldwide began to be impacted by information regarding the COVID-19 Pandemic in February 20, 2020 (Shu et al., 2021). As such, we decided on February 19 as the last period for our event study in order to isolate the effect of the announcement of the export ban on investor behavior.

Lastly, in order to further enhance the validity of our claim of causality, we decided to include a control group in our event study design. The assumption is that this control group would provide a strong counterfactual for what would have happened to the stock price if the announcement never took place, given similar systemic risk and security characteristics. This assumption essentially argues that the palm oil industry is similar enough to the nickel industry in the way their equities behave and are impacted by global macroeconomic trends such as commodity price fluctuations, inflation, and growth. Therefore, palm oil equities would arguably be an appropriate control group in this event study. A violation of this assumption would be a clear divergence in the trajectory of the stock price of palm oil equities and nickel equities even during the estimation window, prior to the announcement of the ban. Consequently, for our control group to be valid and effective, it is important that nickel equities (treatment group) and palm oil equities (control group) be similar in more than just their daily stock returns, such as being similar in their responses to major exogenous economic factors and industry-specific variables such as commodity price fluctuations.

4.2 Model Specifications

Our event study is conducted in four sequential steps. First, Model 1 estimates normal returns using the market model during the estimation window (2 January 2018 to 1 September 2019). Second, Model 2 calculates abnormal returns during the event window by subtracting predicted returns from actual returns. Third, Model 3 aggregates these abnormal returns into cumulative abnormal returns (CARs) for each firm. Finally, Model 4 tests statistical significance through cross-sectional regression, comparing CARs between treatment and

control groups.

In model 1, we first discuss the nature of the model itself. We have a list of J units $\{1, \dots, J\}$, with each unit representing a publicly traded firms that are either a nickel producer firm (treatment group), a nickel consumer firm (treatment group), or a palm oil firm (control group). We choose palm oil firms because of the following: (1) They provide a valid counterfactual as both industries are Indonesian commodity exporters facing similar systematic market risks, (2) they avoid spillover effects because the palm oil markets are structurally separate from both nickel extraction and nickel processing, (3) they are not impacted by the nickel-specific export ban treatment we are studying.

The time window of the study will take place within $T \in \{1, \dots, T_1\}$, with the total time period of the analysis spanning 518 trading days from January 2, 2018 to February 19, 2020. The estimation window is defined as $\{1, \dots, T_0\}$ covering the period from January 2, 2018 to September 1, 2019, which gives us 400 trading days to estimate normal return parameters before the export ban announcement. Furthermore, the event window is $\{T_0 + 1, \dots, T_1\}$, covering the period from September 2, 2019 to February 19, 2020. In this setup, we define $T_0 = 400$ and $T_1 = 518$. We believe this provides a sufficient window to study the causal effects of the announcement by observing the market's response to the policy shock, while terminating before February 20, 2020 to avoid contamination from COVID-19-related market disruptions.

In model 1, for each firm i , $R_{i,t}$ represents the daily stock return at time t during the estimation window. Let $\mathbf{R}_1$ be a $(T_0 \times J_1)$ matrix that collects the daily returns for nickel companies (treatment group) during the estimation window, where each column in $\mathbf{R}_1 = (R_{1,1}, R_{1,2}, \dots, R_{1,T_0})$ corresponds to the return data for a different nickel firms over time. Similarly, let $\mathbf{R}_0$ be a $(T_0 \times J_0)$ matrix that collects the daily returns for palm oil companies (control group), where each column in $\mathbf{R}_0 = (R_{0,1}, R_{0,2}, \dots, R_{0,T_0})$.

The variable $R_{m,t}$ represents the Jakarta Composite Index return at time t , which serves as the market benchmark to control for overall market movements that could confound

the relationship between individual stock returns and the export ban announcement. The firm-specific intercept is denoted as α_i , which captures the component of return that is independent of market movements. The systematic risk coefficient is denoted as β_i , which measures the sensitivity of firm i 's returns to overall market movements.

Using the dataset outlined in Table 1, we estimate the market model using ordinary least squares (OLS) regression in R during the estimation window. The coefficients of interest are α_i and β_i for each firm. These parameters establish each firm's "normal" return relationship with the broader Indonesian market, providing the baseline against which we measure abnormal returns during the event window.

The market model for each individual company's daily stock return is shown below:

Model 1: Market Model (Normal Returns Estimation)

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t} \quad (1)$$

Variables:

- $R_{i,t}$: Daily return of stock i at time t
- α_i : Firm-specific intercept (abnormal return when market return is zero)
- β_i : Systematic risk coefficient for stock i (sensitivity to market movements)
- $R_{m,t}$: Jakarta Composite Index (JCI) return at time t (market return)
- $\varepsilon_{i,t}$: Error term (idiosyncratic shock to stock i at time t)

Using the estimated parameters $\hat{\alpha}_i$ and $\hat{\beta}_i$ from Model 1, we calculate abnormal returns for each company during the event window. For each company i at time t , the abnormal return $AR_{i,t}$ represents the difference between the actual return and the return predicted by the market model. This calculation isolates the component of stock return that cannot be explained by overall market movements, thereby capturing the firm-specific impact of the

export ban announcement. We compute abnormal returns for all trading days in the event window from September 2, 2019 to February 19, 2020.

The abnormal return calculation for each individual company's daily stock return is shown below:

Model 2: Abnormal Returns Calculation

$$AR_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t}) \quad (2)$$

Variables:

- $AR_{i,t}$: Abnormal return for stock i at time t (actual return minus predicted return)
- $R_{i,t}$: Actual daily return of stock i at time t
- $\hat{\alpha}_i$: Estimated firm-specific intercept from Model 1
- $\hat{\beta}_i$: Estimated systematic risk coefficient from Model 1
- $R_{m,t}$: Market return at time t

We aggregate the daily abnormal returns calculated in Model 2 to measure the total impact of the export ban announcement over the entire event window. For each company i , the cumulative abnormal return CAR_i represents the sum of all abnormal returns from the announcement date (September 2, 2019) through the end of the event window (February 19, 2020). This cumulative measure captures the market's complete reassessment of firm value following the policy shock, providing a single metric for the total abnormal return experienced by each company during the event period.

The cumulative abnormal return calculation for each individual company is shown below:

Model 3: Cumulative Abnormal Returns

$$CAR_i[t_1, t_2] = \sum_{t=t_1}^{t_2} AR_{i,t} \quad (3)$$

Variables:

- $CAR_i[t_1, t_2]$: Cumulative abnormal return for stock i over event window from day t_1 to day t_2
- t_1 : Start date of event window
- t_2 : End date of event window
- $AR_{i,t}$: Abnormal return for stock i at time t (from Model 2)

Lastly, for model 4, for each firm i , CAR_i represents the cumulative abnormal return over the entire event window $[T_0 + 1, T_1]$. Let $\mathbf{CAR}_1$ be a $(J_1 \times 1)$ vector that collects the cumulative abnormal returns for nickel companies (treatment group), where each element corresponds to the total abnormal return for a different nickel company. Similarly, let $\mathbf{CAR}_0$ be a $(J_0 \times 1)$ vector that collects the cumulative abnormal returns for palm oil companies (control group).

The variable $Producer_i$ is an indicator variable equal to 1 if firm i is a nickel producer and 0 otherwise. The variable $Consumer_i$ is an indicator variable equal to 1 if firm i is a nickel consumer and 0 otherwise. Palm oil companies serve as the omitted category (baseline group) in this specification.

Using the dataset of firm-level $CARs$, we estimate the cross-sectional regression model using OLS in R. The coefficients of interest are γ_1 and γ_2 , representing the differential $CARs$ for producers and consumers, respectively. These coefficients capture the additional effect of the export ban announcement on the market valuation of nickel companies compared to palm oil companies.

The cross-sectional regression model for cumulative abnormal returns is shown below:

Model 4: Cross-Sectional Regression (Statistical Significance Testing)

$$CAR_i = \gamma_0 + \gamma_1 Producer_i + \gamma_2 Consumer_i + \nu_i \quad (4)$$

Variables:

- CAR_i : Cumulative abnormal return for stock i over the event window
- γ_0 : Intercept (average CAR for control group: palm oil companies)
- γ_1 : Coefficient on producer indicator (differential CAR for producer vs. control)
- $Producer_i$: Indicator variable equal to 1 if firm i is a nickel producer, 0 otherwise
- γ_2 : Coefficient on consumer indicator (differential CAR for consumer vs. control)
- $Consumer_i$: Indicator variable equal to 1 if firm i is a nickel consumer, 0 otherwise
- ν_i : Error term for firm i

5 Results

Using RStudio to run the models and regression specifications, we attain parameter estimate and cumulative daily stock return tables as well as regression tables containing our desired results. We utilize the stargazer function to produce these tables, whilst clustering the standard errors at the firm level.

5.1 Market Model Estimates (Model 1)

Model 1 established the baseline relationship between each firm's stock returns and the broader movements of the Jakarta Stock Exchange Composite Index which represents the market, specifically during the estimation window of January 2, 2018 to September 1, 2019.

The coefficient of interest in this model is β_i , representing the measure of a firm's systematic market risk (an indicator of the price volatility of a stock or other asset in comparison with the broader market JCI) (Kenton, 2025).

Table 5: Market Model Parameter Estimates

Ticker	Company Group	β	p-value	R^2	N
INCO	Nickel Producer	1.462***	0.0000	0.220	400
ANTM	Nickel Producer	1.503***	0.0000	0.232	400
DKFT	Nickel Producer	0.067	0.6574	0.001	399
CITA	Nickel Producer	-0.277	0.4359	0.002	309
ZINC	Nickel Producer	0.339***	0.0074	0.018	400
HRUM	Nickel Consumer	1.314***	0.0000	0.152	400
INAI	Nickel Consumer	0.513***	0.0005	0.030	390
GDST	Nickel Consumer	0.366	0.1791	0.005	398
AALI	Palm Oil (Control)	0.525***	0.0000	0.056	400
SMAR	Palm Oil (Control)	-0.186	0.5822	0.002	156
LSIP	Palm Oil (Control)	0.762***	0.0000	0.068	400
SIMP	Palm Oil (Control)	0.539***	0.0000	0.058	400
TBLA	Palm Oil (Control)	0.786***	0.0000	0.079	400
SGRO	Palm Oil (Control)	0.039	0.7126	0.000	352
DSNG	Palm Oil (Control)	0.147	0.2858	0.003	400

Notes: β measures systematic market risk. R^2 indicates proportion of return variation explained by market movements. ***p<0.01, **p<0.05, *p<0.10.

As seen in Table 5 above, the results of our analysis suggest that there is significant heterogeneity in market risk exposure among the firms even within the same firm sector group. For instance, there is a notable difference in the impact of broader market movements among the different nickel producers, where INCO and ANTM display higher systemic risk, with statistically significant β coefficients of 1.462 and 1.503 at the 1% level, respectively,

while smaller nickel producing firms such as DKFT and CITA, with β coefficients of 0.067 and -0.277 , respectively, barely move with the market, albeit with statistically insignificant results. This indicates that some larger firms display higher systemic risk aligning with the market while smaller cap nickel producing firms may have greater idiosyncratic volatility, potentially as a result of being small-cap equities in an emerging market such as Indonesia.

Furthermore, nickel consumers also seem to have some systematic risk, with HRUM showing strong market sensitivity with a statistically significant β coefficient of 1.314 at the 1% level. Similarly, the palm oil firm which acts as the control group indicate some sensitivity to market movements with β coefficients typically ranging around approximately 0.6 with statistically significant estimates.

The generally low R^2 values seem to suggest that market movements in these industries in the Indonesian equity markets have low explanatory power and high idiosyncratic volatility. These R^2 values range between 1% and 23%, with many falling below 5%. With these results serving as the baseline, one can then carry out calculations to obtain the abnormal returns and thereafter the cumulative abnormal returns required for our event study, specifically in relation to the announcement of regulation 11/2019 of the export ban on nickel ores.

5.2 Cumulative Abnormal Returns (Model 3)

Model 3 sums up the daily abnormal returns calculated in Model 2 to measure the total market effect of the announcement of regulation 11/2019 across the days within the event window. The coefficient of interest in this set of analysis is the CAR_i coefficient, which represents the cumulative abnormal return for each firm during the event window of September 2, 2019 and February 19, 2020.

Table 6: Event Study Results: Cumulative Abnormal Returns by Company Group

Company Group	N	Mean CAR (%)	Median CAR (%)	SD (%)	Min CAR (%)	Max CAR (%)	t-statistic	Sig.
Nickel Producer	5	-45.14	-58.18	29.82	-78.97	-8.13	-3.39	**
Nickel Consumer	3	-5.46	-7.60	8.47	-12.65	3.88	-1.12	
Palm Oil (Control)	7	10.24	14.17	16.97	-16.38	36.13	1.60	

Notes: Cumulative abnormal returns (CAR) over event window September 2, 2019–February 19, 2020 (118 trading days). T-statistics test null hypothesis of zero mean CAR. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6 shows the mean CAR_i by firm sector group. As shown above, the results indicate that there is a statistically significant negative effect of the nickel export ban on nickel producers. On average, nickel producers experienced a -45.14% decrease in its CAR_i , at the 5% level. This may suggest that the markets viewed raw nickel ore extraction in a downward trajectory following the announcement of regulation 11/2019 as compared to what the market would have perceived nickel producers in the counterfactual. The median CAR_i is even higher, which may indicate that the effect of the announcement was even more significant for the typical nickel extractor.

In contrast, nickel consumers saw a smaller decline in its cumulative abnormal return that is statistically insignificant ($CAR_i = -5.46\%$, $p > 0.10$). Therefore, it can be concluded that both the upside and downside effect of the announcement on market reactions is limited.

A similar conclusion can be drawn from the results of the palm oil analysis. The control group showed a positive CAR_i of 10.24% , albeit statistically insignificant. This positive return among the palm oil companies is in support of the overall event study design, as palm oil seems to be unimpacted by market reactions to the announcement of the nickel export ban.

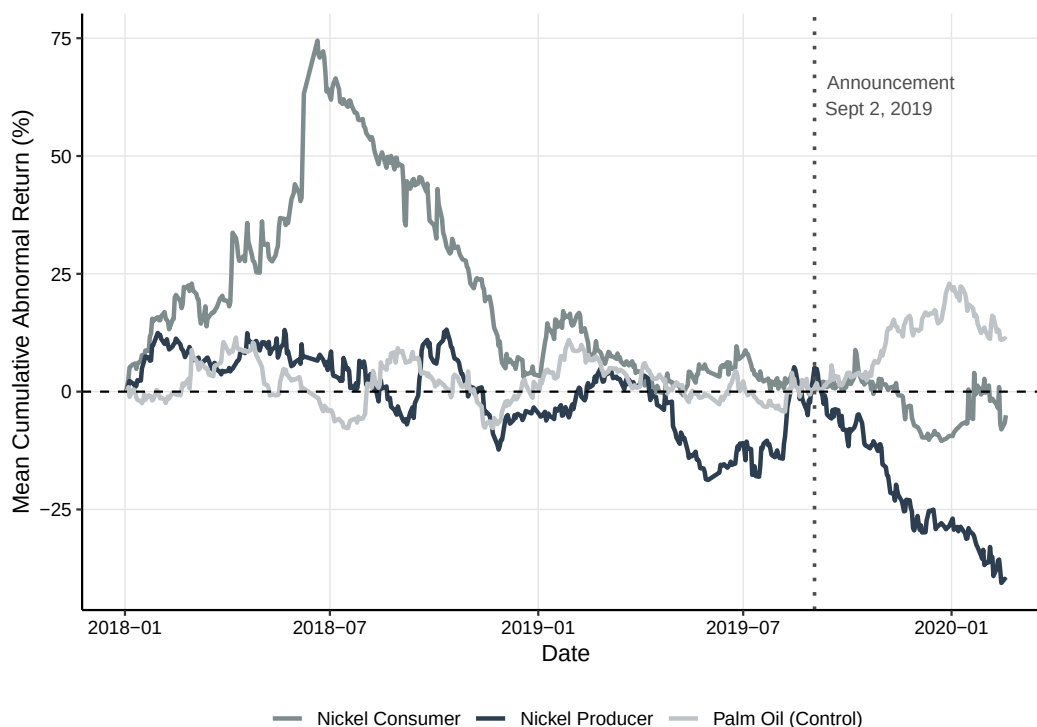


Figure 1: Cumulative Abnormal Returns Over Event Window by Company Group

As shown in Figure 1, the movement of the cumulative abnormal returns across time corroborates the hypothesis of this paper that the impact of the nickel export ban announcement would negatively affect nickel producer cumulative abnormal returns. This temporal dynamics of the market reaction can be seen when cumulative abnormal returns are calculated across the entire study period of January 2018 to February 2020. During the estimation window of January 2018 to September 2019, we can see that all three firm industry groups show cumulative abnormal returns that either fluctuate around zero or experience positive growth, which we believe is further evidence that both the treatment and control groups follow similar enough daily stock return trends prior to the announcement. In the aftermath of the nickel export ban on September 2, 2019, one can clearly see that the cumulative abnormal returns for nickel producers break away from the trajectory of the other groups quite drastically in a negative direction. This direction suggests that the markets quickly incorporated this newly announced information about the export ban's direct negative impact

on nickel producers' ability to generate revenue and profit. In contrast, one does not see a similar impact on investor reactions when it comes to nickel consumer equities, signaling that investors may not perceive a similar negative impact on nickel consumers, especially since the intent of this ban was to support the growth of such industries domestically. It is important to note that nickel consumer equities also do not see any growth in cumulative abnormal returns. These findings strongly suggest that the market saw the export ban as a negative factor in growth of value for nickel producing firms, which may be an unintended consequence of the government's aim to foster a domestic industry as firms at the forefront of the supply chain may be detrimentally impacted.

5.3 Cross-Sectional Regression (Model 4)

Lastly, Model 4 established the causal and statistical relationship of the treatment effects, that is of the announcement, through cross-sectional regression of CAR_i on indicator variables that represent each firm industry group. The coefficients of interest are γ_1 and γ_2 , which measure the differential cumulative abnormal returns for nickel producers and consumers, respectively, relative to the palm oil control group.

Table 7: Cross-Sectional Regression: Treatment Effects on Cumulative Abnormal Returns

<i>Dependent variable: CAR (%)</i>	
Producer Indicator	−55.377*** (12.454)
Consumer Indicator	−15.691 (14.677)
Constant	10.236 (8.039)
Control Group	Palm Oil Companies
Event Window	Sep 2, 2019–Feb 19, 2020
Observations	15
R ²	0.626
Adjusted R ²	0.563
Residual Std. Error	21.269 (df = 12)
F Statistic	10.031*** (df = 2; 12)
<i>Notes: *p<0.1; **p<0.05; ***p<0.01.</i>	

We can see in Table 7 above that the nickel producer indicator coefficient estimate is −55.377, which suggests that on average nickel producers experience cumulative abnormal returns of 55.38 percentage points lower than that of the control group during the period of the event window. This coefficient is statistically significant at the 1% level, which provides strong statistical evidence that the announcement of regulation 11/2019 has a causal and significantly negative impact on the market’s reaction to nickel producer firm equities, which on average represents up to half of the firms’ value.

On the other hand, the nickel consumer indicator coefficient estimate is −15.691, which suggests that nickel consumers experienced returns of 15.69% lower when compared to the control group palm oil. However, this coefficient is not statistically significant at even the

10% level, potentially due to the limited sample size of nickel consumers in the Indonesian equity markets as of early 2020. This statistical insignificance may also mean that the markets did not have a reaction that is meaningfully different to its position on palm oil firm equities, suggesting that there is no causal impact of the announcement of regulation 11/2019 on nickel consumer equities either in a negative nor the intended positive way.

Lastly, the constant term of 10.236 represents the mean cumulative abnormal return for the palm oil firms which acts as the control group. As seen in the R^2 , the model explains 62.6% of the variation in the CAR_i , with an adjusted R^2 of 0.563. Given that the F-Statistic of 10.031 is statistically significant at the 1% level, the overall model fit achieves the desired performance. These regression results confirm the causal relationship that was observed as a pattern in the results of model 3, which provides strong evidence in support of our hypothesis: that the announcement of regulation 11/2019 banning the export of raw nickel ores would have a detrimental causal effect on how the markets price nickel producer equities, which may render the effectiveness of the policy limited in the short term.

5.4 Limitations

First, the small sample size utilized in this study may be the primary limitation in producing more statistically significant results. Given the nature of the nascent nickel industry and Indonesia as an emerging market with yet to be established data collection practices, we were only able to obtain the daily stock returns of 5 nickel producers and 3 nickel consumers with sufficient data availability. Consequently, this may be the primary cause of the limited statistical power of the results obtained to detect the heterogeneous effects on cumulative abnormal returns. These results are therefore driven by the few large nickel firms such as INCO and ANTM, given many other potentially insightful firms such as LION had to be excluded due to the severity of missing data.

Second, the appropriateness of palm oil as a comparable control group to the nickel firms may be challenged as well. It is a rather challenging task to identify a similar export

commodities industry established enough in Indonesia to have sufficient daily stock returns. Although arguably similar enough, the patterns of commodity price fluctuations and volatility that ultimately drive the perceived value of equities in this industry is not a one-to-one match. There are also plenty of other structural differences and unobserved factors that drive these two industries to have different trajectories.

Lastly, the short event window utilized in this study, which was done out of necessity to isolate the impact of the nickel export ban announcement from other factors contaminating the study such as the outbreak of the pandemic, may prevent us from capturing the long-term effects of the announcement.

6 Conclusion

In summary, we conclude that announcement of the Regulation 11/2019, namely the nickel export ban, has an adverse effect on nickel producers who relied heavily on such export markets for revenue. This regulation on raw nickel exports had the initial aim of encouraging domestic value addition to bolster the domestic nickel processing industry through export restriction. The policy was intended to increase the profitability for both domestic nickel producers and consumers. However, it has not achieved both goals, while severely impacting the market's position on nickel producers especially. The decreasing cumulative abnormal returns of nickel producing firms is contrasted by the stable growth of palm oil firms and the stagnant growth of nickel consuming firms. This divergence in the trajectory suggests that in the absence of such an export ban announcement, the trajectory of cumulative abnormal returns of nickel producing firms would not be such a downward direction.

Indeed, the results of the study reveals a nuanced finding: although the long-term benefits of this policy have yet to be seen, the short-term pains felt by the nickel industry is real. This underscores the challenges that governments face when attempting to utilize trade-restrictive policies to foster domestic industries and grow the economy. Future researchers should delve

into the possibility of avoiding the financial drawbacks of such policies whilst still encouraging the growth of local industries. In the long-run, the level of success of the policy may increase as the markets incorporate new and potentially more positive information with respect to both nickel producing firms and nickel consuming firms, especially as the industry further adjusts to this new regulation. Regardless, policymakers must take into account both the long-run benefits and short-run drawbacks of the export ban in order to formulate better regulations in the pursuit of economic growth.

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A Appendix

A.1 Detailed Market Model Regression Tables

Table 8: Market Model Results - Nickel Producers

	<i>Dependent variable:</i>				
	Stock Return (%)				
	INCO	ANTM	DKFT	CITA	ZINC
	(1)	(2)	(3)	(4)	(5)
Constant	1.462*** (0.138)	1.503*** (0.137)	0.067 (0.150)	-0.277 (0.355)	0.339*** (0.126)
Market Return	0.039 (0.126)	0.131 (0.125)	-0.095 (0.137)	0.212 (0.318)	0.159 (0.115)
Observations	400	400	399	309	400
R ²	0.220	0.232	0.001	0.002	0.018

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 9: Market Model Results - Nickel Consumers

	<i>Dependent variable:</i>		
	Stock Return (%)		
	HRUM IJ Equity	INAI IJ Equity	GDST IJ Equity
	(1)	(2)	(3)
Constant	1.3139*** (0.1553)	0.5126*** (0.1471)	0.3663 (0.2722)
Market Return	-0.0930 (0.1418)	0.0457 (0.1351)	0.0295 (0.2473)
Observations	400	390	398
R ²	0.1524	0.0303	0.0046

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 10: Market Model Results - Palm Oil Control

<i>Dependent variable:</i>							
	Stock Return (%)						
	AALI	SMAR	LSIP	SIMP	TBLA	SGRO	DSNG
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Constant	0.525*** (0.108)	-0.186 (0.337)	0.762*** (0.141)	0.539*** (0.109)	0.786*** (0.135)	0.039 (0.105)	0.147 (0.137)
Market Return	-0.058 (0.099)	-0.007 (0.304)	-0.024 (0.129)	-0.094 (0.100)	-0.072 (0.123)	-0.035 (0.097)	-0.065 (0.125)
Observations	400	156	400	400	400	352	400
R ²	0.056	0.002	0.068	0.058	0.079	0.000	0.003

Note:

*p<0.1; **p<0.05; ***p<0.01

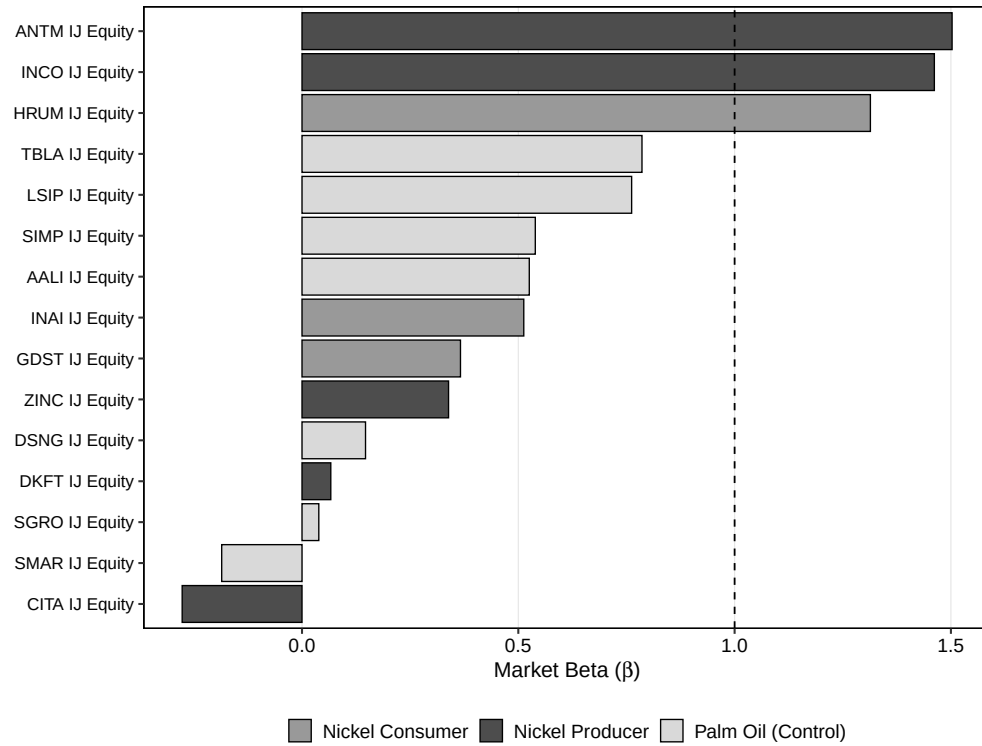


Figure 2: Market Model Beta Coefficients by Company Group

A.2 Abnormal Returns Summary Statistics

Table 11: Abnormal Returns Summary Statistics by Company

Ticker	Company Group	Mean AR	SD AR	Min AR	Max AR
INCO	Nickel Producer	−0.170	2.930	−5.606	7.878
ANTM	Nickel Producer	−0.497	2.043	−4.723	6.381
DKFT	Nickel Producer	−0.675	3.731	−10.173	13.619
CITA	Nickel Producer	−0.070	3.550	−16.406	10.865
ZINC	Nickel Producer	−0.517	2.854	−16.755	6.836
HRUM	Nickel Consumer	0.033	1.343	−4.252	5.188
INAI	Nickel Consumer	−0.079	3.747	−20.074	16.274
GDST	Nickel Consumer	−0.108	5.113	−17.097	29.146
AALI	Palm Oil (Control)	0.136	2.135	−6.882	5.810
SMAR	Palm Oil (Control)	−0.161	2.344	−6.784	8.489
LSIP	Palm Oil (Control)	0.032	2.324	−5.016	6.047
SIMP	Palm Oil (Control)	0.175	1.890	−3.871	6.878
TBLA	Palm Oil (Control)	−0.019	1.829	−4.122	5.666
SGRO	Palm Oil (Control)	0.159	2.528	−7.456	6.626
DSNG	Palm Oil (Control)	0.308	1.752	−4.569	6.414

Notes: All values in percentage points. Abnormal returns (AR) calculated as actual returns minus predicted returns from market model over event window (September 2, 2019–February 19, 2020). Statistics based on 117 trading days for most companies.

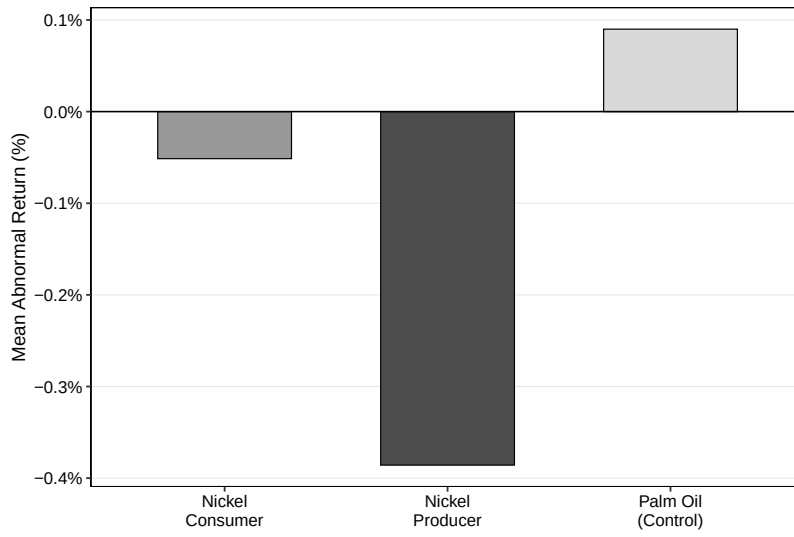


Figure 3: Mean Abnormal Returns by Company Group During Event Window

A.3 Individual Company Cumulative Abnormal Returns

Table 12: Cumulative Abnormal Returns by Individual Company

Ticker	Company Group	CAR (%)	Valid Days	Missing (%)
INCO	Nickel Producer	−19.92	117	0.85
ANTM	Nickel Producer	−58.18	117	0.85
DKFT	Nickel Producer	−78.97	117	0.85
CITA	Nickel Producer	−8.13	117	0.85
ZINC	Nickel Producer	−60.50	117	0.85
HRUM	Nickel Consumer	3.88	117	0.85
INAI	Nickel Consumer	−7.60	96	18.64
GDST	Nickel Consumer	−12.65	117	0.85
AALI	Palm Oil (Control)	15.87	117	0.85
SMAR	Palm Oil (Control)	−16.38	102	13.56
LSIP	Palm Oil (Control)	3.70	117	0.85
SIMP	Palm Oil (Control)	20.42	117	0.85
TBLA	Palm Oil (Control)	−2.27	117	0.85
SGRO	Palm Oil (Control)	14.17	89	24.58
DSNG	Palm Oil (Control)	36.13	117	0.85

Notes: CAR calculated over event window September 2, 2019–February 19, 2020.

Valid Days indicates number of trading days with non-missing return data. Missing (%) shows percentage of days with missing data. LION IJ Equity excluded from analysis due to 41% missing data in event window.

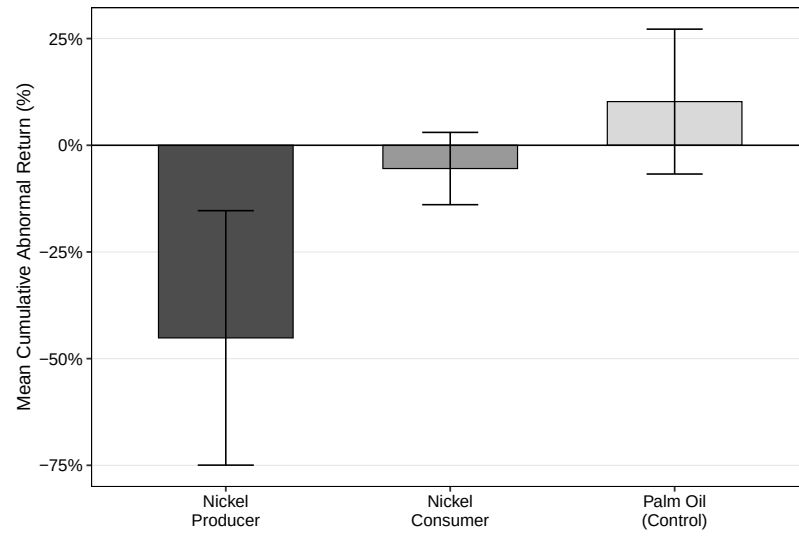


Figure 4: Mean Cumulative Abnormal Returns by Company Group